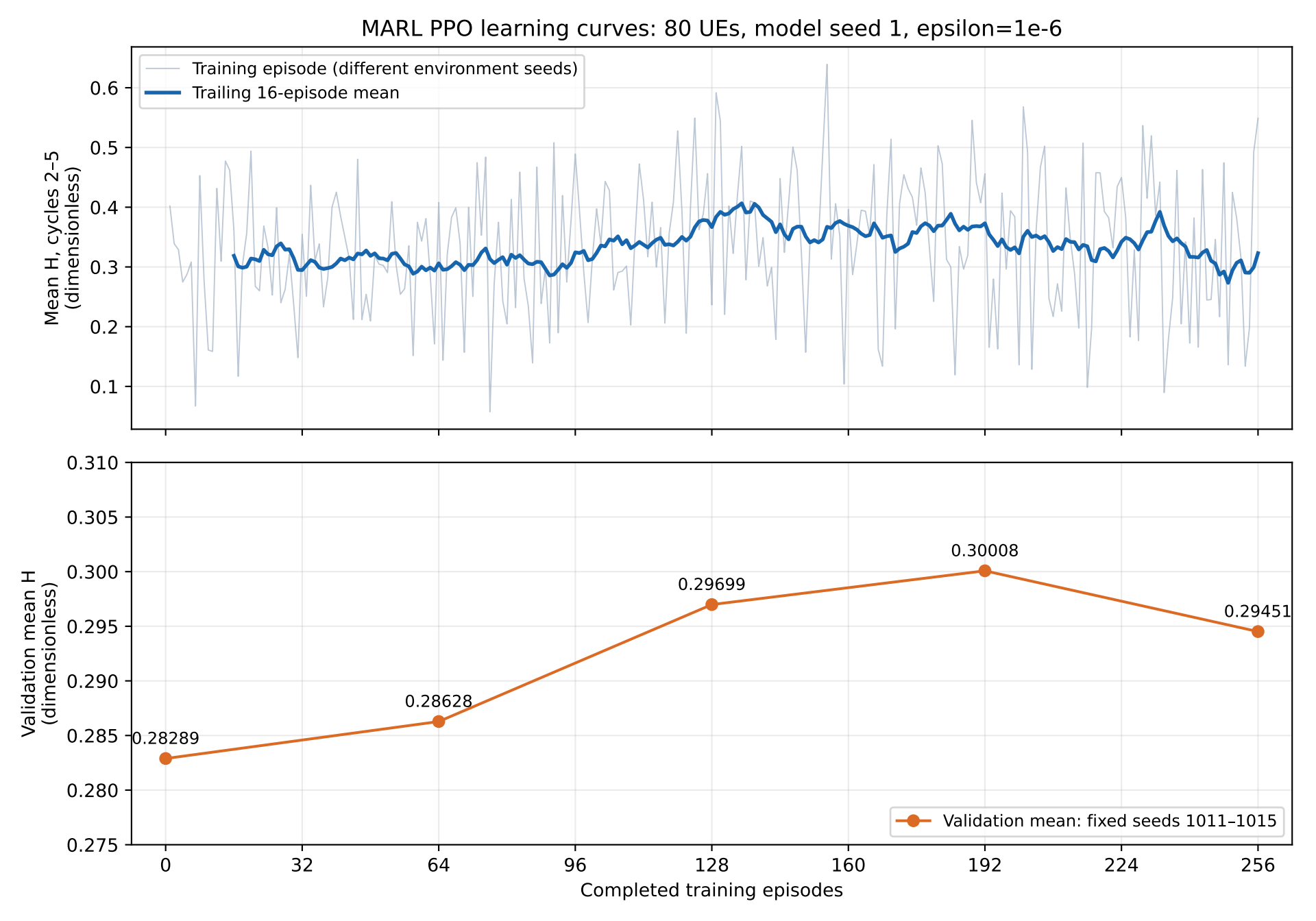
MARL PPO learning curves: 80 UEs, model seed 1, epsilon=1e-6
Training episode (different environment seeds)
Trailing 16-episode mean
Mean H, cycles 2–5 (dimensionless)
0.6
0.5
0.4
0.3
0.2
0.1
Validation mean H (dimensionless)
0.310
0.305
0.300
0.295
0.290
0.285
0.280
0.275
0.28289
0.28628
0.29699
0.30008
0.29451
Validation mean: fixed seeds 1011–1015
0
32
64
96
128
160
192
224
256
Completed training episodes